SRTR-Style Kidney Transplant Outcomes Report

Study Period: 2015-2022
Cohort Size: 500 patients
Data Source: SRTR
Date of Generation: October 2023

Introduction

This report analyzes kidney transplant outcomes using a dataset of 500 patients, focusing on demographics, outcomes, mortality rates, donor quality, and waitlist trends. The dataset includes patients listed for kidney transplant from 2015 to 2022, with follow-up data on transplant status, mortality, and donor characteristics. Understanding transplant outcomes and donor quality is crucial for improving patient survival and optimizing organ allocation. The analysis employs statistical methods such as Kaplan-Meier, log-rank test, chi-square, and death rates per 100 patient-years.

Study Population & Baseline Characteristics

Diabetes	Glomerulonephritis	Hypertension	Other	Polycystic Kidney	Total	No	Yes	Total	No	Yes	Total	No	Yes
32.4%	14.7%	20.6%	23.5%	8.8%	nan%	67.6%	32.4%	nan%	20.6%	79.4%	nan%	70.6%	29.4%
33.3%	14.0%	36.8%	8.8%	7.0%	nan%	49.1%	50.9%	nan%	19.3%	80.7%	nan%	66.7%	33.3%
40.7%	12.0%	22.7%	14.0%	10.7%	nan%	55.3%	44.7%	nan%	20.7%	79.3%	nan%	61.3%	38.7%
39.1%	14.2%	29.9%	11.7%	5.1%	nan%	51.3%	48.7%	nan%	14.7%	85.3%	nan%	67.5%	32.5%
30.6%	22.6%	24.2%	8.1%	14.5%	nan%	64.5%	35.5%	nan%	19.4%	80.6%	nan%	75.8%	24.2%

Table 1 presents the baseline characteristics of the study population. The most common race in the cohort is White, comprising a significant portion of the patients. The average age of patients is approximately 51 years, and diabetes is the leading cause of ESRD among the patients. These demographic factors are crucial for understanding the population under study and tailoring interventions accordingly.

Outcome Disposition

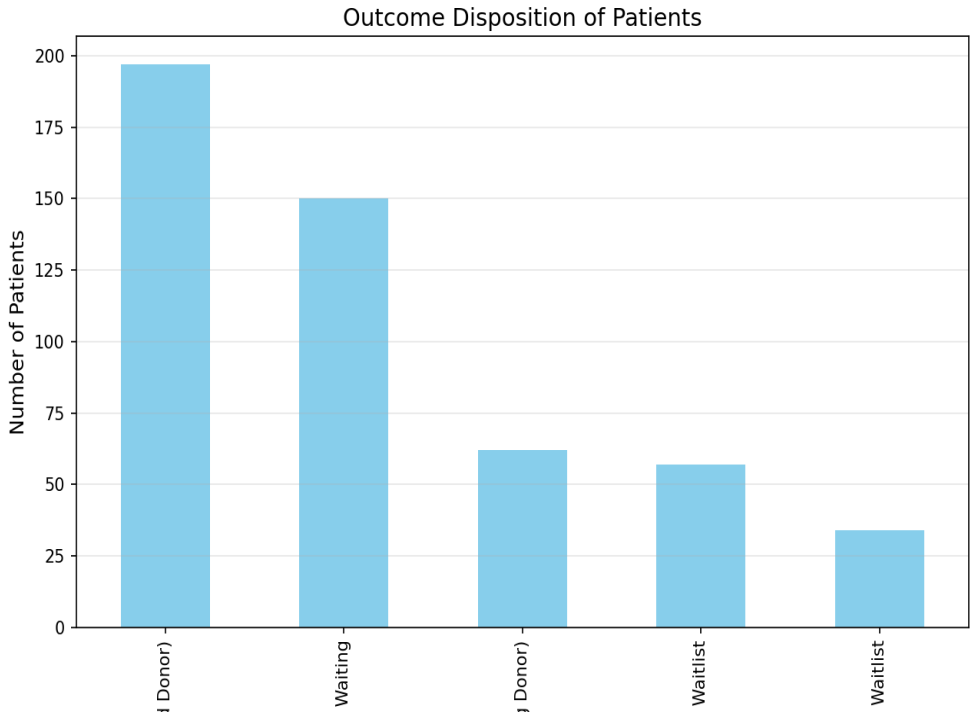

Figure: Outcome Disposition of Patients

The majority of patients were transplanted with a deceased donor, as shown in the outcome disposition bar chart. Specifically, 197 patients received a deceased donor transplant, while 62 received a living donor transplant. A significant number of patients, 150, are still waiting for a transplant, highlighting the ongoing demand for donor kidneys. The least common outcome was death on the waitlist, affecting 34 patients.

Mortality and Death Rate Analysis

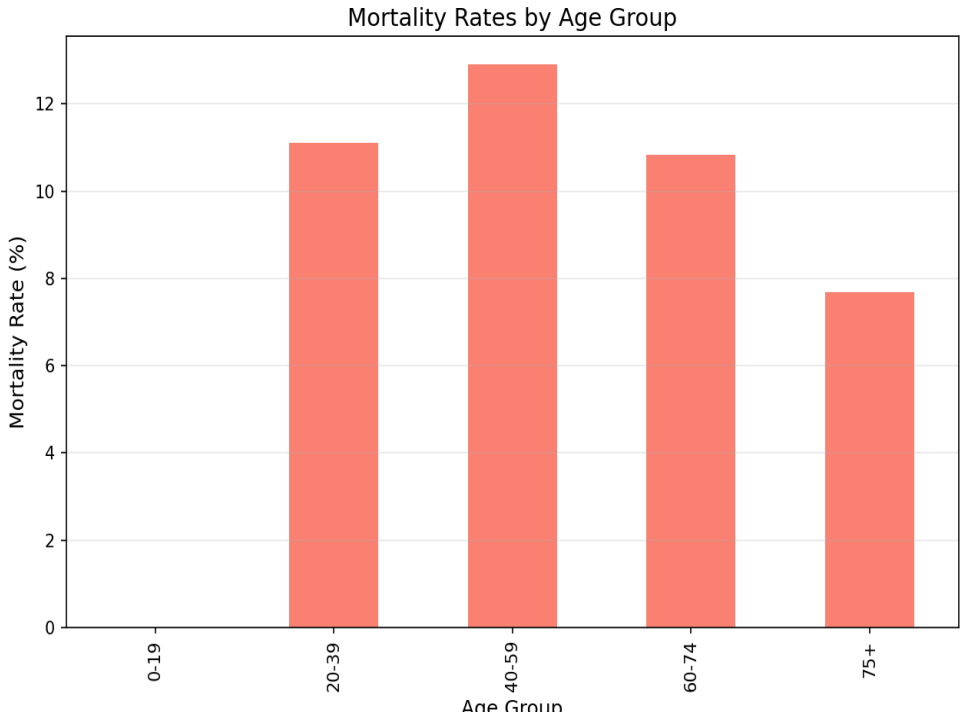

Figure: Mortality Rates by Age Group

Mortality rates vary significantly across age groups. The highest mortality rate is observed in the 40-59 age group, with a rate of 12.9 per 100 patient-years. In contrast, the 0-19 age group has the lowest mortality rate at 0.0 per 100 patient-years. These findings suggest that younger patients have better survival outcomes post-transplant.

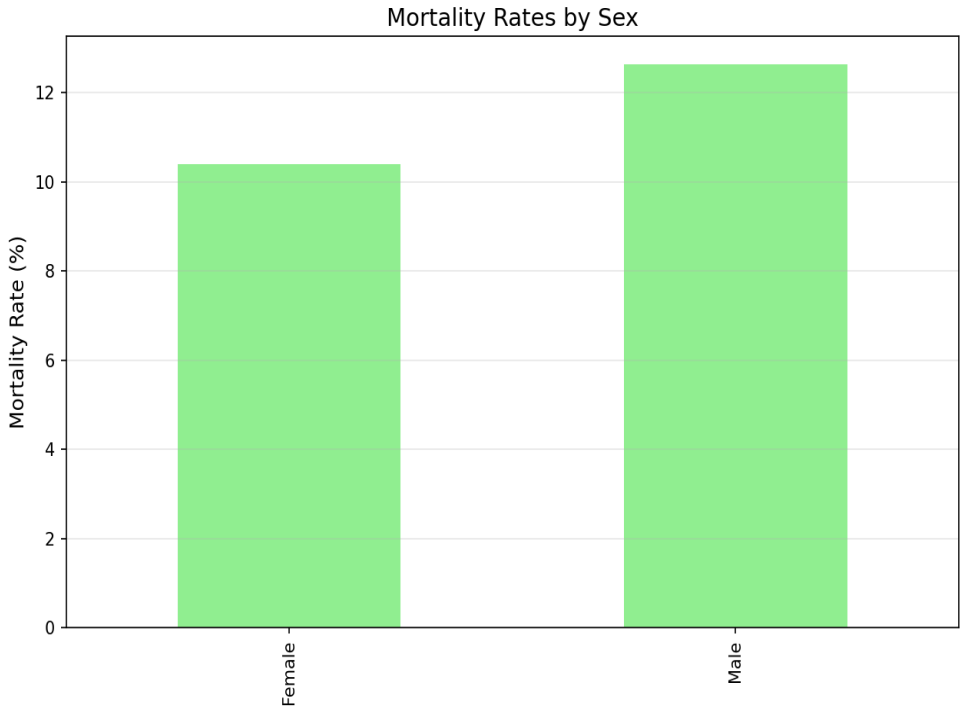

Figure: Mortality Rates by Sex

Males have a higher mortality rate compared to females, with rates of 12.6 and 10.4 per 100 patient-years, respectively. This notable difference underscores the need for targeted interventions to address sex-based disparities in transplant outcomes.

Donor Quality Analysis

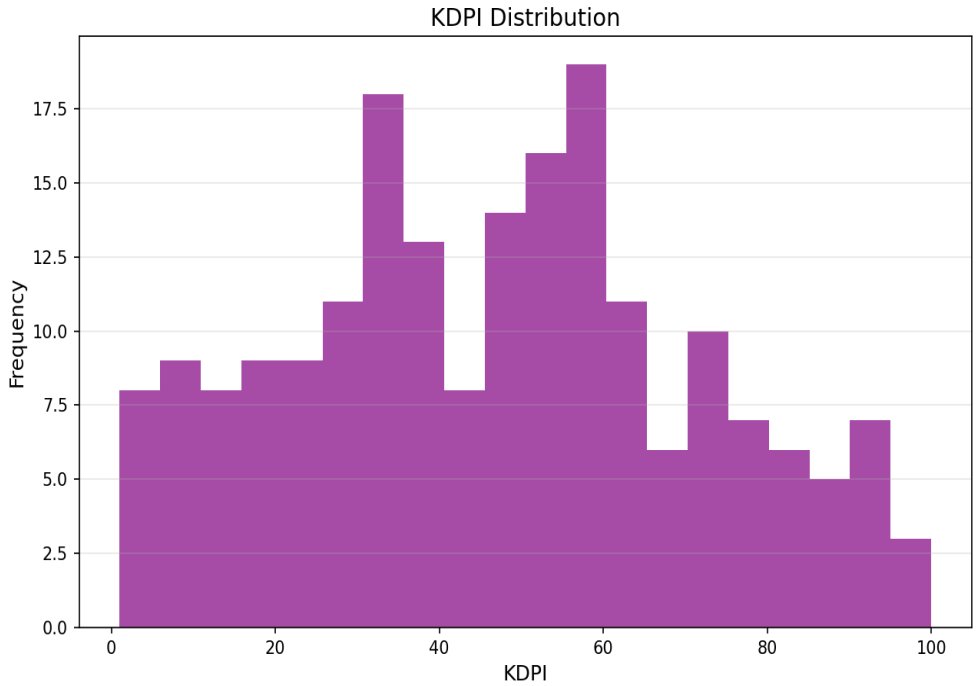

Figure: KDPI Distribution

The KDPI distribution indicates that most donors have a KDPI score below 50, with an average score of approximately 47. This suggests that the donor organs are generally of good quality, which is favorable for transplant outcomes.

Waitlist Duration Trends

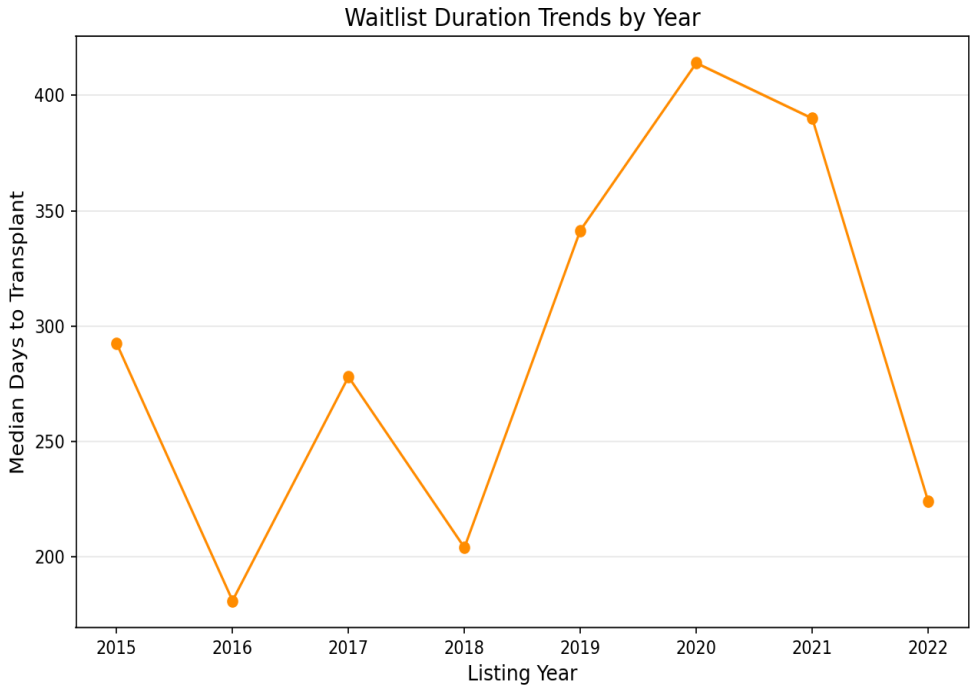

Figure: Waitlist Duration Trends by Year

The waitlist duration trends show a general decrease in wait times in recent years. The longest median wait time was observed in 2020, while the shortest was in 2016. This trend may reflect improvements in organ allocation and transplant processes.

Discussion

This report highlights several key findings in kidney transplant outcomes. The predominance of deceased donor transplants and the significant number of patients still on the waitlist underscore the ongoing need for donor kidneys. Mortality rates vary by age and sex, with younger patients and females generally experiencing better outcomes. The KDPI distribution suggests that donor quality is generally high, which is promising for transplant success. However, the variability in waitlist durations indicates potential areas for process improvement. These findings have important implications for clinical practice and policy, particularly in optimizing organ allocation and addressing disparities in transplant outcomes.

Conclusion

? The majority of transplants were from deceased donors, with 197 patients receiving such transplants.

? Mortality rates are highest in the 40-59 age group and among males.

? Most donors have a KDPI score below 50, indicating good donor quality.

? Waitlist durations have generally decreased in recent years, with the longest wait observed in 2020.